



SIMATS
ENGINEERING



SIMATS
Sreeetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Cloud-Based Smart Warehouse Automation

Simulation-Based Intelligent Warehouse Automation System

CSA1501 cloud computing

Guide By:

Dr.BOOMIJA

SSE, SIMATS Engineering

Presented by:

H. Harshavardhan (192511171)

Ch . Abhilash Reddy (192525462)

Lingamguntla surya praneeth(192311233)

Yannamani sarvarayudu(192324181)

SSE, SIMATS Engineering

ABSTRACT

- A cloud-deployable warehouse automation platform combining a FastAPI/PostgreSQL backend with a full multi-robot fleet simulation and a 2D Digital Twin.
- Real-time A* path planning with obstacle avoidance, cell-reservation collision detection, and priority-based conflict resolution drives autonomous robot navigation.
- An explainable AI layer forecasts demand (WAPE/MAE/RMSE/sMAPE with walk-forward validation) and flags inventory shrinkage using Isolation Forest anomaly detection.
- A Scenario Lab lets managers run reproducible, seeded what-if experiments — including OR-Tools optimized scheduling and SimPy discrete-event packing simulations.
- Security is enforced through JWT/RBAC authentication, step-up OTP verification, and a hash-chained, tamper-evident audit ledger.
- System health, metrics, and error tracking are exposed through Prometheus, Grafana, and Sentry for full observability.

INTRODUCTION

- Manual and semi-automated warehouses struggle with robot congestion, unpredictable demand, and undetected inventory shrinkage.
- This project builds a single integrated platform where a simulated robot fleet, inventory intelligence, and warehouse security work together end to end.
- A tick-based Digital Twin keeps a live, database-reconciled view of every robot, task, and obstacle in the warehouse.
- Machine learning adds forecasting and anomaly detection on top of raw stock-movement data, so decisions are explainable, not black-box.
- The system is built to be demoable and auditable: every simulation run is seeded and reproducible, every security-relevant action is logged to an immutable ledger.

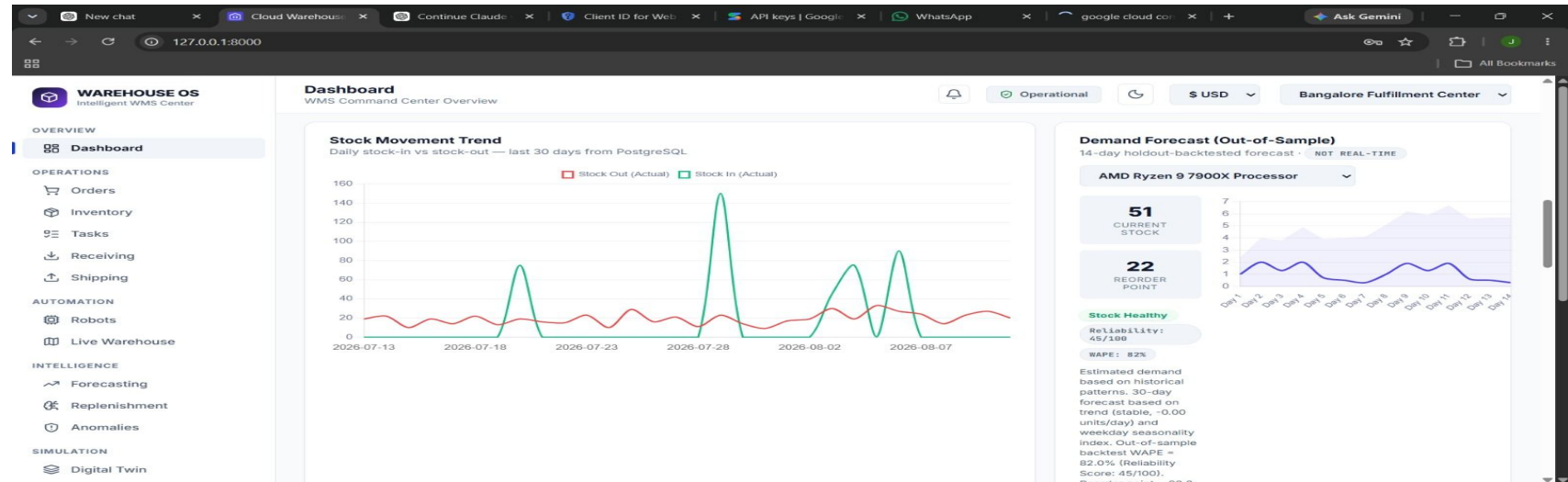
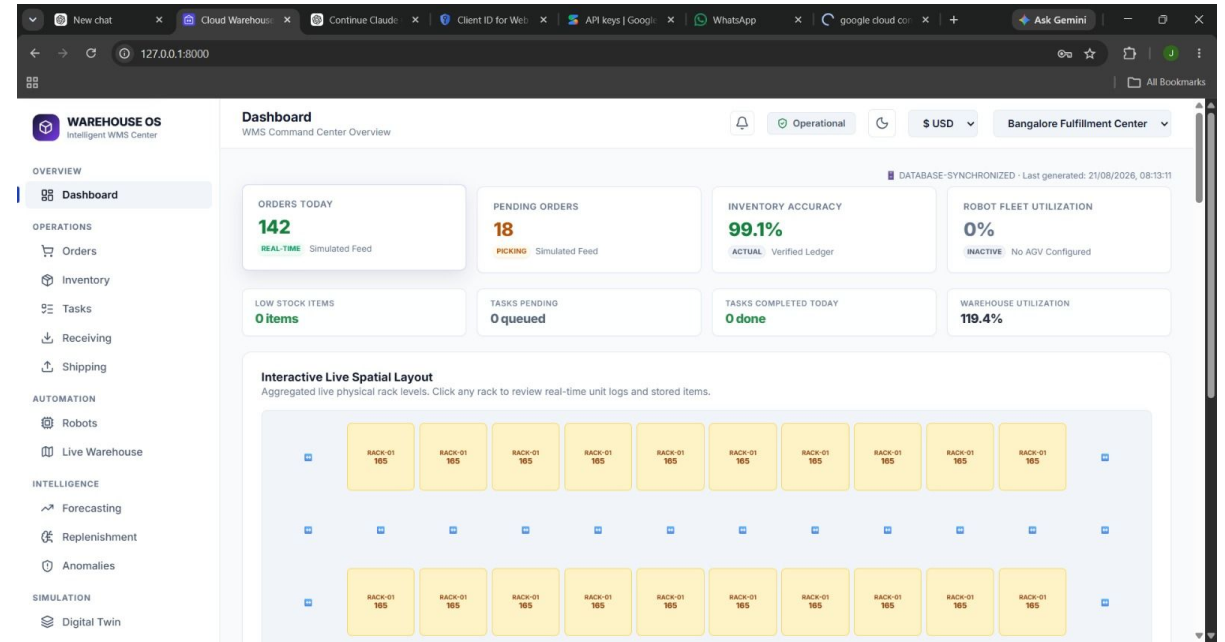
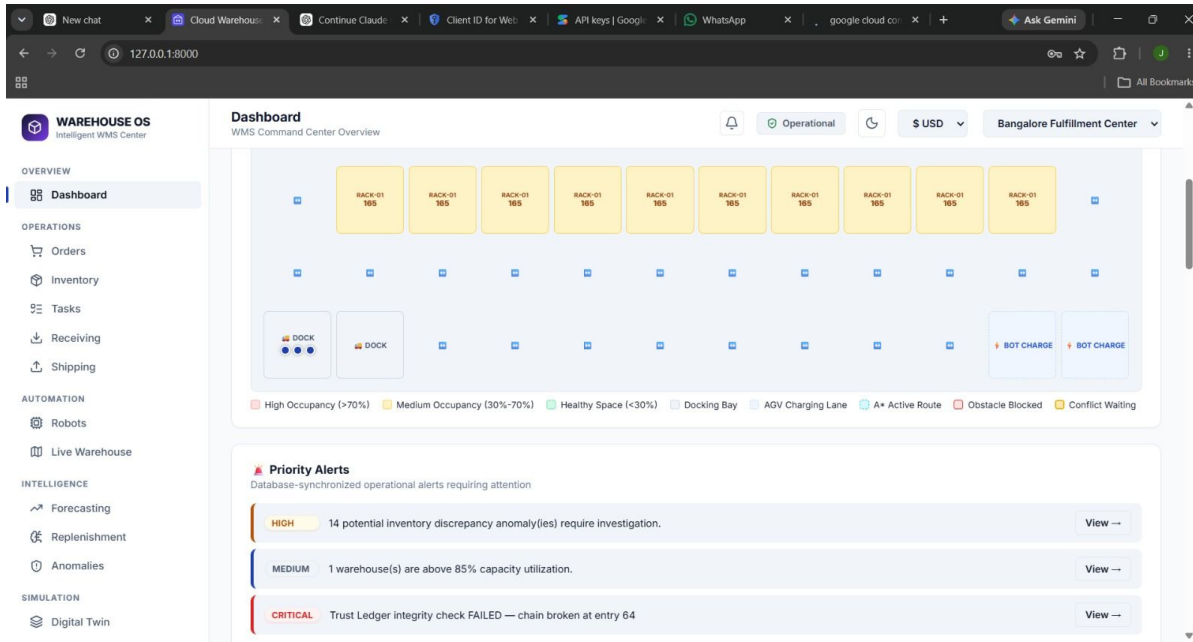
OBJECTIVES

- To simulate a multi-robot warehouse fleet with A* path planning, dynamic replanning, and collision-free navigation.
- To build a 2D Digital Twin that visualizes robot state, routes, obstacles, and warehouse KPIs in real time.
- To forecast product demand and detect inventory anomalies using explainable machine learning models.
- To secure the platform end to end with RBAC, step-up authentication, and a hash-chained audit ledger, while exposing full observability through metrics and error tracking.

Module 1: Multi-Robot Fleet Simulation & Path Planning

- Implements A* pathfinding with a Manhattan-distance heuristic over a weighted, obstacle-aware warehouse grid.
- Detects and resolves same-cell and head-on collisions between robots using a tick-indexed reservation table.
- Applies priority-based conflict resolution so lower-priority robots yield and wait rather than collide.
- Triggers dynamic replanning when a robot is blocked for multiple ticks, generating a fresh route around the obstruction.
- Assigns tasks to robots using cost-based, battery-aware scoring so low-charge robots are not routed onto long jobs.

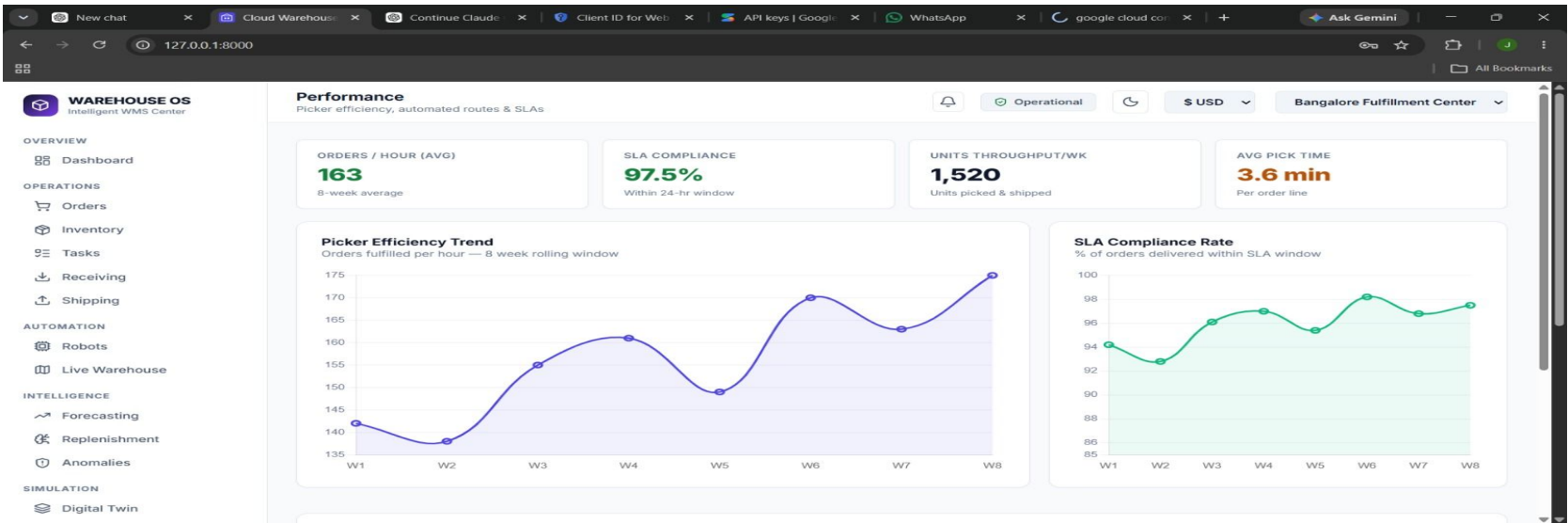
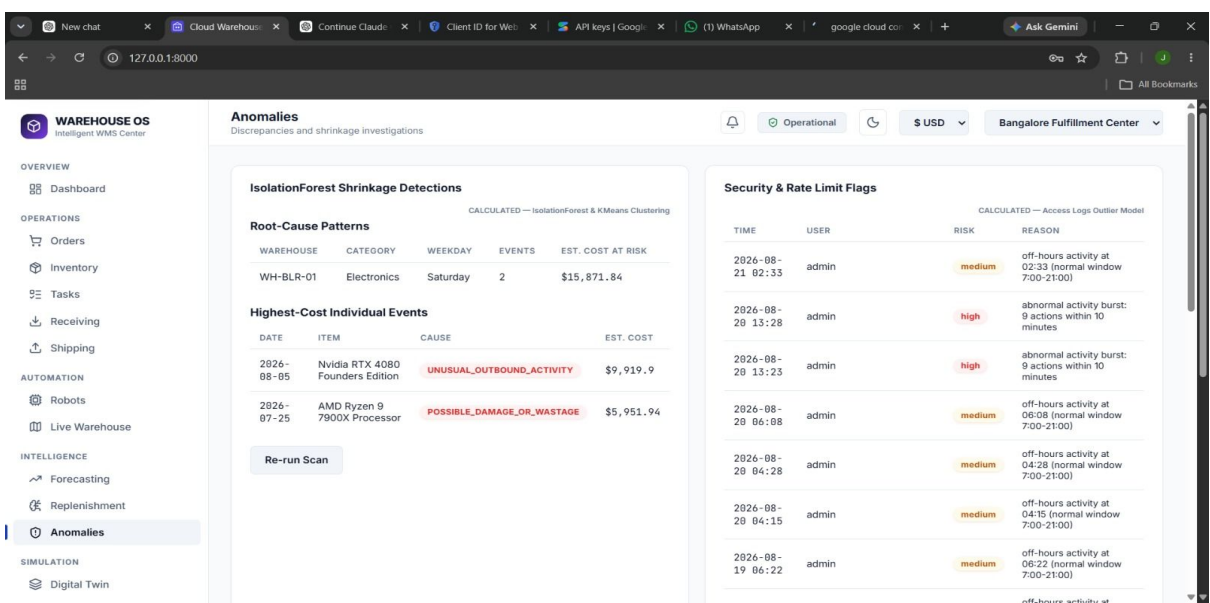
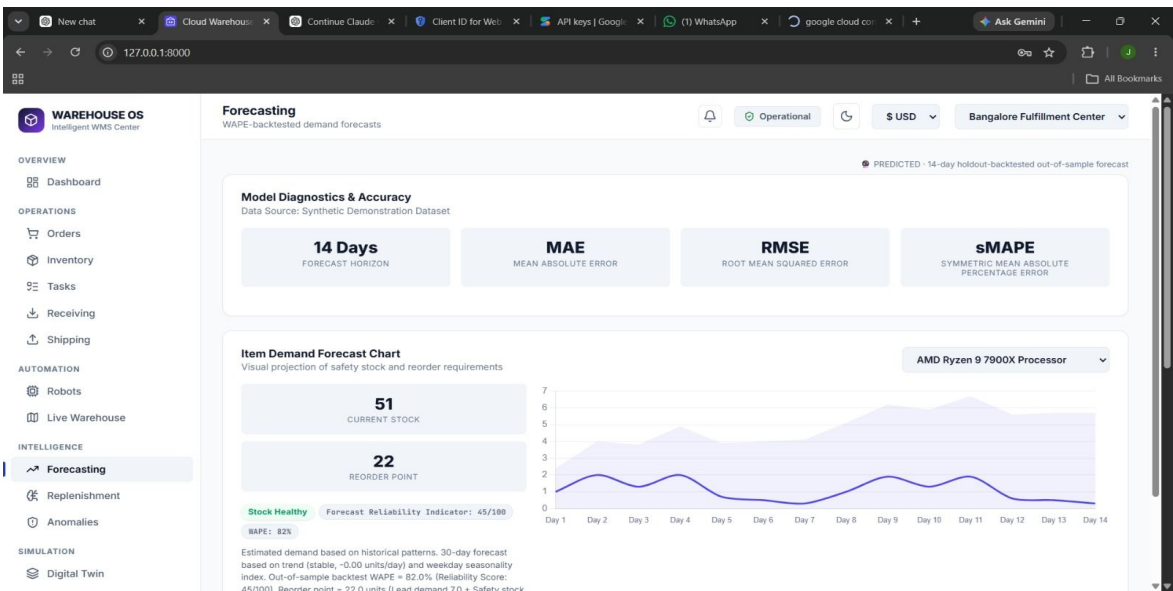
Module 1 : Output



Module 2: Demand Forecasting & Explainable Anomaly Detection

- Forecasts short-horizon product demand using chronological train/test splits and walk-forward validation against naive and moving-average baselines.
- Scores forecast quality with WAPE, MAE, RMSE, and sMAPE, and attaches a composite reliability score to every forecast.
- Runs an Isolation Forest model over stock-movement data to flag inventory anomalies that may indicate shrinkage, theft, or damage.
- Performs ABC inventory classification to prioritize which items need the closest monitoring and fastest replenishment.
- Surfaces every AI recommendation with a plain-language explanation and a human approval step before it is acted on.

Module 2: Output



Module 3: 2D Digital Twin & Scenario Lab

- Maintains a database-reconciled 2D Digital Twin showing live robot positions, routes, obstacles, and warehouse KPIs.
- Provides Start / Pause / Step / Resume / Reset simulation controls, all wired to real backend endpoints.
- Runs reproducible, seed-based What-If experiments through a dedicated Scenario Lab, with full experiment history and result export.
- Benchmarks robot-task assignment using Google OR-Tools (CP-SAT) against the warehouse's real, live robot fleet.
- Models packing-station throughput and operator contention using a discrete-event SimPy simulation.

Module 4: Security, Audit & Observability

- Enforces Role-Based Access Control with granular permissions, bcrypt password hashing, and JWT-based sessions.
- Requires step-up OTP verification for sensitive actions, with account lockout after repeated failed login attempts.
- Logs every security- and inventory-relevant event to a hash-chained, tamper-evident audit ledger with built-in integrity verification.
- Exposes real-time application metrics via Prometheus and visualizes them on Grafana dashboards.
- Tracks application errors and performance through Sentry, with a system health endpoint reporting live status of every connected service.

Module 4: Output

The screenshot shows the 'System Health' dashboard for WAREHOUSE OS. The left sidebar contains navigation links for Overview, Operations, Automation, Intelligence, and Simulation. The main content area is titled 'System Health' and includes a 'System Health' status indicator (Operational) and a 'Bangalore Fulfillment Center' dropdown. The dashboard displays various system metrics and their operational parameters.

SERVICE SUBSYSTEM	STATUS BADGE	MEASURED LATENCY	OPERATIONAL PARAMETERS & MESSAGE
PostgreSQL Database	HEALTHY	1.5 ms	Schema Revision: c6a0f47c242e Pool Size: 10
Redis Cache	UNAVAILABLE	-	Failed to retrieve Redis client connection
RabbitMQ Message Broker	HEALTHY	-	DLQ Depth: 0 billing log: 0
Celery Task Workers	UNAVAILABLE	-	No active workers detected on Celery control plane
SMTP / Resend Email	HEALTHY	-	Alert email SMTP settings present
Sentry Client SDK	HEALTHY	-	Sentry client integration is active
OpenAI Cognitive Client	HEALTHY	-	Gemini API is configured (gemini-3.5-flash-lite)
Cloudflare Proxy Edge	NOT CONFIG	-	CLOUDFLARE_API_TOKEN environment variable missing

The screenshot shows the 'Audit Log' dashboard for WAREHOUSE OS. The left sidebar contains navigation links for Overview, Operations, Automation, Intelligence, and Simulation. The main content area is titled 'Audit Log' and includes a 'Bangalore Fulfillment Center' dropdown. The dashboard displays a 'Ledger Chain Compromised' alert and a table of audit log entries.

BLOCK ID	TIMESTAMP	EVENT / ACTION	DETAILS & METADATA	SHA-256 HASH	PREVIOUS HASH
#1084	21/08/2026, 03:55:29	NOTIF PUBLISHED USER LOGIN	Event Type: USER_LOGIN Warehouse Id: null Severity: INFO Source Entity Id: 1 Source Entity Type: USER Recipients Count: 3	fac4d31d...d28781eb	fe16c789...dbcc12d5
#1083	21/08/2026, 03:55:22	USER LOGIN	Username: admin Role: admin Method: password Ip: 127.0.0.1 Time: 2026-08-21T03:55:22.072674	fe16c789...dbcc12d5	7dee15c2...e9b1db92
#1082	21/08/2026, 03:55:14	NOTIF PUBLISHED USER LOGIN	Event Type: USER_LOGIN Warehouse Id: null Severity: INFO Source Entity Id: 1 Source Entity Type: USER Recipients Count: 3	7dee15c2...e9b1db92	c1c02624...4ed1b8f9
#1081	21/08/2026, 03:55:14	USER LOGIN	Username: admin Role: admin Method: password	7dee15c2...e9b1db92	c1c02624...4ed1b8f9

The screenshot shows the 'Users & Roles' dashboard for WAREHOUSE OS. The left sidebar contains navigation links for Overview, Operations, Automation, Intelligence, and Simulation. The main content area is titled 'Users & Roles' and includes a 'Bangalore Fulfillment Center' dropdown. The dashboard displays a table of users and their roles, along with a 'Permissions Matrix' section.

USERNAME	FULL NAME	ROLE	STATUS	LOGIN METHOD	LAST ACTIVITY	ACTIONS
test_admin_hardened	test_admin_hardened	ADMIN	Active	password	19/08/2026, 18:29:13	Deactivate
test_admin	test_admin	ADMIN	Active	recovery	19/08/2026, 18:29:39	Deactivate
test_manager	test_manager	MANAGER	Active	password	19/08/2026, 18:20:02	Deactivate
test_viewer	test_viewer	VIEWER	Active	password	20/08/2026, 13:29:11	Deactivate
admin	System Administrator	ADMIN	Active	password	21/08/2026, 03:55:22	Deactivate

Permissions Matrix
RBAC capability overview by role level

CAPABILITY	ADMIN	MANAGER	OPERATOR	VIEWER
Create Warehouses / Items	✓	✓	✗	✗

PROJECT COMPONENT REQUIREMENTS

Hardware Requirements

- Intel Core i5 Processor or higher
- 8 GB RAM
- 256 GB SSD
- Laptop/Desktop with Internet Connection
- Cloud Database (PostgreSQL, Render-hosted)

Core Libraries

- SQLAlchemy, Alembic
- scikit-learn, pandas, NumPy
- Google OR-Tools, SimPy

Software Requirements

- Windows 10/11
- Python 3.11+
- Visual Studio Code
- FastAPI Framework
- PostgreSQL / SQLite (testing only)

Dataset Requirements

- Synthetic Warehouse Inventory Dataset
- Multi-Warehouse Stock-Movement Logs
- Simulated Robot Telemetry & Task Events

TASKS

- Design the warehouse grid, robot fleet model, and A* path-planning engine with collision detection.
- Build the FastAPI backend, PostgreSQL schema, and Alembic migrations for orders, tasks, robots, and inventory.
- Train and validate the demand-forecasting and shrinkage-detection models, and expose them through backend endpoints.
- Build the 2D Digital Twin, Scenario Lab, and AI Decision Center frontend, wiring every module end to end.
- Implement RBAC, OTP, audit ledger, and observability (Prometheus, Grafana, Sentry), and validate with automated tests.

SPECIFIC PARAMETERS

- **Robot State:** Position, battery level, task assignment, and status (idle / moving / waiting / failed), tracked per tick.
- **Path Cost:** Weighted grid cost combined with the Manhattan heuristic to compute each robot's route.
- **Product Demand:** Historical stock-movement volume used as the basis for the forecast model.
- **Inventory Deviation:** Difference between expected and actual stock levels, used to flag anomaly events.
- **Task Priority:** Composite score combining order urgency, robot cost, and battery level for assignment.

OUTCOME PARAMETERS

- **Route & Collision Report:** Shows generated robot routes and any conflicts resolved through waiting or replanning.
- **Forecast Reliability Score:** Displays demand forecast accuracy against naive and moving-average baselines.
- **Shrinkage Risk Classification:** Classifies flagged anomalies as likely shrinkage, damage, or data error.
- **Scenario Comparison Report:** Compares baseline vs. OR-Tools optimized scheduling across repeated simulation runs.
- **Audit & System Health Digest:** Summarizes ledger integrity status and live connection status of every integrated service.

TECHNIQUES TO BE USED

- **Robotics & Path Planning:** A* search, Manhattan heuristic, cell-reservation collision avoidance.
- **Machine Learning:** Isolation Forest for anomaly detection; walk-forward validated demand forecasting.
- **Optimization & Simulation:** Google OR-Tools (CP-SAT) scheduling and SimPy discrete-event simulation.
- **Backend & Data:** FastAPI for the API layer, SQLAlchemy/PostgreSQL for persistence, Alembic for migrations.
- **Security & Observability:** RBAC, OTP step-up auth, hash-chained audit ledger, Prometheus, Grafana, Sentry.

FUTURE SCOPE

- Integrate real IoT/RFID sensors for live robot telemetry instead of simulated data.
- Deploy the async task pipeline on live Redis and RabbitMQ infrastructure for true background processing at scale.
- Improve forecast accuracy using deep-learning models trained on larger historical datasets.
- Extend the Digital Twin from 2D to a full 3D warehouse visualization.
- Add a native mobile app for warehouse managers with push notifications for critical alerts.

CONCLUSION

- The platform simulates a full multi-robot warehouse fleet with real-time, collision-free path planning.
- It forecasts demand and detects shrinkage using explainable, validated machine learning rather than black-box alerts.
- The Scenario Lab enables reproducible, seed-based experimentation and optimization comparison.
- Security and observability are built in from the ground up, not bolted on: RBAC, audit ledger, metrics, and error tracking all work end to end.
- Overall, the project delivers a practical, auditable, and reproducible simulation platform for intelligent warehouse automation.

REFERENCES

- Bodne, M., Patar, V., Sontakke, S., Turankar, V., & Kale, B. V. (2026). Cloud-Based Inventory Management System: Architecture, Real-Time Analytics, Automation, and Security for Modern Enterprise Supply Chains. *International Research Journal of Innovations in Engineering and Technology*, 10(5), 241-253.
- Bhargav, A., Zope, A. S., Jambhale, M. S., Korde, N. M., & Khan, H. A. (2017). IoT Based Industrial Automation. *International Journal of Engineering Research and Technology*, 5(1), 821-827.
- Olanrewaju, R. F. (2021). Cloud-Based Warehouse System for Effective Management of Under and Over-stock Hazards. *ICCCE*, pp. 274-278.
- Tan, W. C., & Sidhu, M. S. (2022). Review of RFID and IoT Integration in Supply Chain Management. *Operations Research Perspectives*, 9, 1342-1348.

Thank You